MILITARY SPECIFICATION

ANTIPYRINE AND BENZOCAINE SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Antipyrine and Benzocaine Solution (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be in accordance with the tests, standards, and requirements of the N.F. XIII, including any supplements or revisions thereto, and shall be in accordance with the following formula:

	Per 15 ml
Glycerin, USP (anhydrous)	17.90 Grams
Antipyrine	0.81 Grams
Benzocaine, NF	0.21 Grams
8 Hydroxyquinoline Sulfate	0.0057 Grams

3.1.1 Moisture. Shall contain not more than 0.2 percent moisture (water) at the time of delivery to the Government, and not more than 0.6 percent moisture for the remainder of the storage life of 3 years.

3.2 Ingredients.

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3.2.1 Glycerin. The Glycerin used in the manufacture of the solution, shall be in accordance with the tests, standards and requirements of the USP including any supplements or revisions thereto.

3.2.2 Benzocaine. The Benzocaine used in the manufacture of the solution, shall be in accordance with the tests, standards and requirements of the NF including any supplements or revisions thereto.

3.2.3 Antipyrine. The Antipyrine used in the manufacture of the solution shall be in accordance with the tests, standards and requirements of the NF XIII including any supplements or revisions thereto.

3.2.4 8-Hydroxyquinoline sulfate. The 8-Hydroxyquinoline sulfate shall be in accordance with the following:

3.2.4.1 Melting range. Shall be 175° C ± 2°C when determined by the USP method for Class I.

3.2.4.2 Assay. Shall assay to contain 74.2 percent ± 0.5 percent of 8-Hydroxyquinoline, when tested as specified in 4.5.1.

3.3 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the solution, to the date of delivery to the Government.

3.4 Workmanship. The material and its containers shall be free from the defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration

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dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container.
Minor	II	2.5	Filled immediate container.

4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

Table II. Raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level*	Minimum sample
Moisture	3.1.1	S-2	5
Ingredients: 8-Hydroxyquinoline sulfate	3.2.4	**	-

* Where possible, the sample shall be used for two or more tests.
** Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

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4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Categories and defects*

Major
101 Solution not clear and free from particles.
102 Immediate container not supplied with dropper closure or accompanying sterile dropper.

Minor
201 Exterior of immediate container is not free from contents or foreign matter.
202 Label not free from defacing marks and tears.

* Inspection is not restricted to the classified possible defects listed above.

4.4.2 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units. For labeling the acceptance number shall be zero.

Table III. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

NOTE: The above examination is only applicable to the unit, intermediate and exterior containers.

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4.5 Tests.

4.5.1 8-Hydroxyquinoline Sulfate

Reagents.

0.1N sodium thiosulfate.

10% potassium iodide. Dissolve 10 gram of potassium iodide in about 80 ml of purified water. Bring to volume of 100 ml with purified water.

Starch T.S. Triturate 1 gram of starch with 10 ml of cold purified water and pour slowly, with constant stirring, into 200 ml of boiling purified water. Boil the mixture until a thin translucent fluid is obtained. Allow to settle and use only the clear supernatant liquid.

Procedure.

Transfer approximately 150 mg of sample, accurately weighed, to a suitable iodine flask. Add 125 ml of dilute hydrochloric acid. Prepare a concomitant reagent blank. Treat sample and blank as follows:

To each flask, add 50 ml of 0.1 N bromine, stopper and mix. Allow to stand for 10 minutes. Add 70 ml of purified water and 10 ml of potassium iodide solution, mix well. Titrate the liberated iodine with 0.1 N sodium thiosulfate using starch T.S. as an indicator.

Calculations.

$$\frac{(V_b - V_s) \times 3.629 \times 100}{W_s} = \% \text{ 8-hydroxyquinoline}$$

Where: V_b = volume of 0.1 N sodium thiosulfate required to titrate the blank

V_s = volume of 0.1 N sodium thiosulfate required to titrate sample

3.629 = mg of 8-hydroxyquinoline equivalent to 1 ml of 0.1 N bromine

W_s = weight of sample in mg.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 15 ml, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available bottle with dropper closure or with accompanying sterile dropper considered acceptable by the Food and Drug Administration (FDA) for the product container therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If a unit package is furnished, each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Thirty-six intermediate packages (432 units) shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Thirty-six intermediate packages (432 units) shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

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5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

Notes: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied the National Stock Number shall appear on the immediate container.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-598-5830	ANTIPYRINE AND BENZOCAINE SOLUTION, 15 ml.

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6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1478

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-A-37336(DSA-DM) ANTIPYRINE AND BENZOCAINE SOLUTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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